Sahaj Bhadja

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Summary

Software Engineer with 2+ years building **distributed platform services**, **data pipelines**, and **resilient cloud workflows** using Python, Java, SQL, Linux tooling, and reliability-focused engineering practices.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Built **scalable backend services** over **Azure SQL/PostgreSQL** for 500+ users, reducing dashboard latency from **1.2s to 350ms** through schema tuning, indexing strategy, and query-path optimization.
- Implemented **fault-aware validation pipelines** for 5K+ monthly jobs using FastAPI orchestration and reconciliation checks, improving **system resilience**, incident triage speed, and production reliability outcomes.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*
- Developed **Python simulation pipelines** processing 100K+ trajectory logs with reproducible runs, enabling performance analysis and rapid iteration across distributed experimentation and workload-evaluation workflows.
- Produced structured validation and debugging documentation, improving cross-team post-mortems, issue isolation, and rollout confidence for model and platform changes in research-to-production handoffs.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Engineered **API and ETL services** using OOP, AWS Glue, and S3, improving response time by **30%** across multi-stage distributed workflows and cross-system integration paths.
- Maintained **20+ unit/integration suites** with Jenkins, PyTest, and Mockito across three OS environments, boosting pipeline efficiency by **68%** and reducing technical debt by **20%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*
- Built **REST API and WebSocket components** supporting eight workflows and 2M+ synchronized records, improving platform throughput and reliability under concurrent operational and data synchronization workloads.
- Automated deployments and testing with **Docker** and Jenkins, reducing deployment time by **50%** and strengthening preventive controls through repeatable release pipelines and environment consistency checks.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Collaborated with product and engineering teams to deliver **Flask + React** features serving 10K+ daily requests, balancing rapid feature iteration with maintainable architecture and code quality standards.
- Added **CI/CD quality controls** and concurrent processing safeguards, reducing turnaround time by **60%** while improving release predictability, testability, and operational consistency under high-volume usage patterns.

Skills

Languages: **Python**, **Java**, SQL, C++, JavaScript/TypeScript
Platform & Systems: Linux workflow, distributed service design, network-aware debugging, performance profiling
Cloud & Infra: Kubernetes, Docker, AWS (S3/Glue/RDS), Azure SQL, CI/CD, Git
Data & Pipelines: PostgreSQL, MySQL, MongoDB, ETL orchestration, telemetry-style validation workflows
Quality & Reliability: Unit/integration testing, Jenkins, PyTest, incident documentation, post-mortem driven improvements

Projects

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*
- Built **distributed platform workflows** for 50K+ users and 1M+ transactions with **Kubernetes**-backed deployment, secure integrations, and resilient data synchronization under high-throughput operational conditions.

Chat2SQL Code | *Java, TypeScript, SQL, Maven, Docker*
- Developed a multi-database analytics engine spanning 10+ databases with **schema-aware query generation** and execution validation, improving reliability, query correctness, and maintainability of data-intensive workflows.

HMIS Live | *MERN, ABDM APIs, Twilio*
- Designed **API-driven operational services** supporting 100+ daily appointments, adding monitoring-backed controls that improved record retrieval by **2x** and strengthened fault diagnosis across production incidents.